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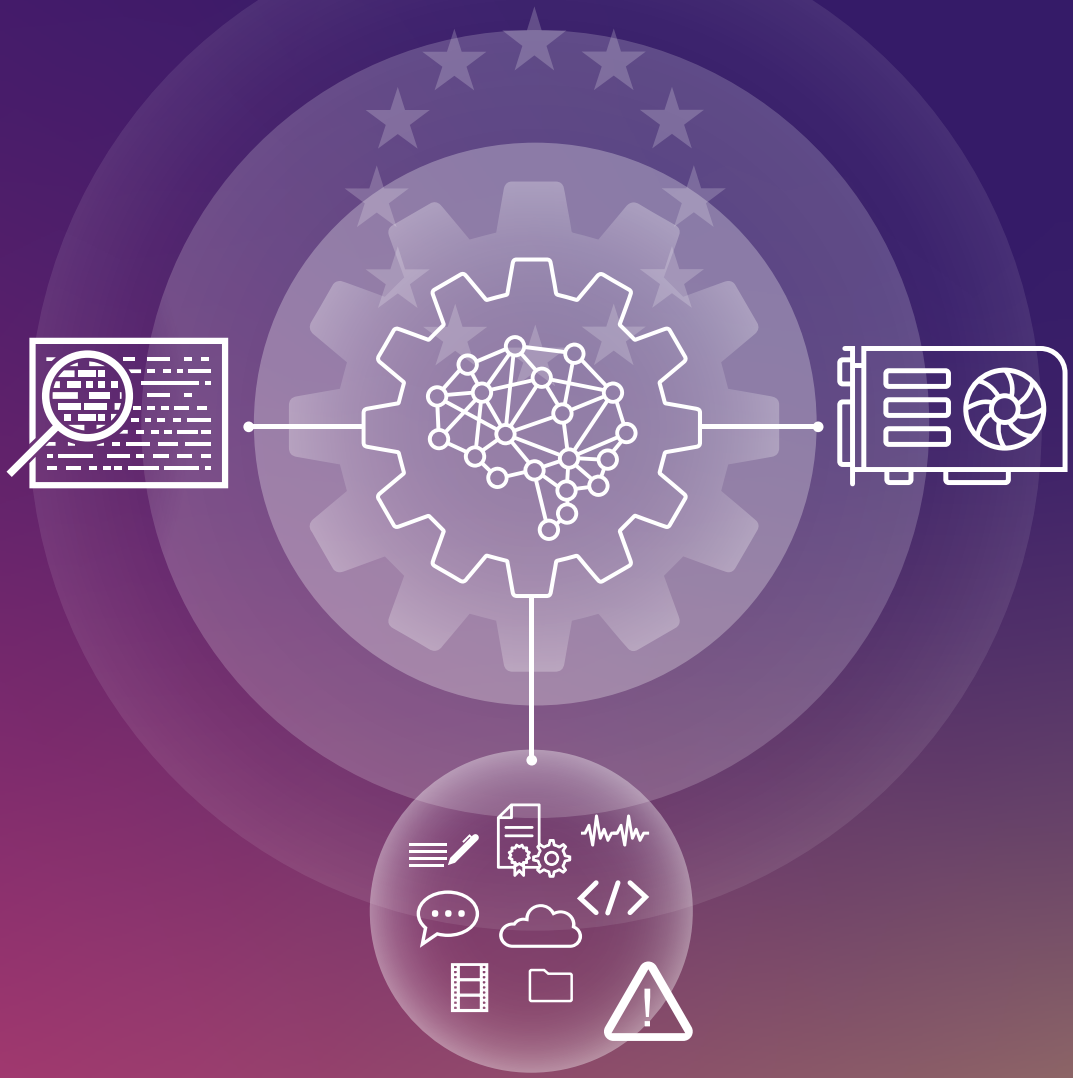
The Role of AI Safety Benchmarks in Evaluating Systemic Risks in General-Purpose AI Models

Collection of External Scientific Studies on General-Purpose AI Models under the EU AI Act

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Abstract

The evaluation of systemic risks in General-Purpose AI (GPAI) models is a complex challenge that requires a multifaceted approach, extending beyond traditional capability assessments. This report analyses the role of AI safety benchmarks in identifying systemic risks in GPAI models, proposing a dual-trigger framework that combines capability triggers with safety benchmarks to provide a more comprehensive assessment of potential harms. The current landscape of safety benchmarks is still in development, with various initiatives emerging to address specific systemic risk categories, such as the ones identified in the GPAI Code of Practice: cyber offence, chemical, biological, radiological and nuclear (CBRN) risks, harmful manipulation, and loss of control. A tiered evaluation strategy is recommended, applying more rigorous and costly safety evaluations only to models that meet a predefined capability threshold or are intended for deployment in high-risk domains, ensuring proportionality and efficient resource allocation. Ultimately, the development of robust and standardised safety benchmarks is relevant for accurate classification of GPAI models as GPAI models with systemic risk, and policy initiatives should incentivise their creation to enable more effective systemic risk identification.

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Note from the Editors

The EU AI Act entered into force on 1 August 2024, with the aim of promoting innovation in and uptake of AI in the Union, while ensuring a high level of protection of health, safety and fundamental rights, including democracy and the rule of law. Chapter V of the AI Act outlines obligations for the providers of general-purpose AI (GPAI) models, which are AI models *"trained with a large amount of data using self-supervision at scale, that [display] significant generality and [are] capable of competently performing a wide range of distinct tasks [...] and that can be integrated into a variety of downstream systems or applications"*. Moreover, the chapter specifies additional obligations for the most advanced GPAI models, those that pose systemic risks, which are classified as such according to criteria established in Article 51 and Annex XIII. From 2 August 2025, the obligations for providers of GPAI models and GPAI models with systemic risk enter into application.

The European Commission's Joint Research Centre (JRC) has been providing scientific support throughout the legislative process of the AI Act since 2020. After the Council and Parliament reached a final agreement in December 2023, the JRC initiated an internal study, which included two external experts, focusing on the technical aspects of Chapter V that likely required further clarification. This study generated an internal report titled "General Purpose AI Models under the AI Act" (Hernández-Orallo et al., 2024), which provided preliminary insights into compute, generality, capabilities, and systemic risks. One of the clear conclusions of this preliminary study was that further scientific work was necessary.

Between September 2024 and June 2025, the JRC setup and managed, in close collaboration with the EU AI Office, a pool of 15 external experts with diverse expertise and backgrounds. This expert pool produced further technical scientific input on key aspects of Chapter V, through the development of methodologies for categorising AI models as GPAI models and for classifying GPAI models as GPAI models with systemic risk, to inform implementation of the EU AI Act. The experts also provided input on the recently published Commission guidelines on the scope of obligations for providers of GPAI models (European Commission, 2025), as part of the public multi-stakeholder consultation. The primary outcome of this expert pool is this **Collection of External Scientific Studies on General-Purpose AI Models under the EU AI Act**, which comprises a total of six external scientific study reports.

Although more documents may be added to the collection in the future, as of the writing of this editorial, the titles of the external reports included in the collection are:

- Training Compute Thresholds - Key Considerations for the EU AI Act
- A Framework for General-Purpose AI Model Categorisation
- A Framework to Categorise Modified General-Purpose AI Models as New Models Based on Behavioural Changes
- A Proposal to Identify High Impact Capabilities of General-Purpose AI Models
- The Role of AI Safety Benchmarks in Evaluating Systemic Risks in General-Purpose AI Model
- General-Purpose AI Model Reach as Criterion for Systemic Risk

The overall objective of this collection is twofold. On the one hand, it aims to contribute to broadening the understanding and discussion of the technical and scientific issues related to GPAI models and the identification of systemic risks. On the other hand, it seeks to provide a solid scientific basis for informing the implementation of Chapter V of the EU AI Act, which has recently entered into application. It is clear that we are dealing with complex issues, where a clear scientific consensus has yet to be established, and which require a certain degree of flexibility. Nevertheless, this is part of the

necessary effort to promote innovation and the uptake of AI, while ensuring protection for human health, safety, and fundamental rights in Europe.

These external scientific studies cover aspects regarding the presumption of having high impact capabilities based on cumulative amount of computation used for training (Article 51(2)), notification conditions (Article 52), the definition of a GPAI model (Article 3(63)), and considerations for GPAI models being classified as GPAI models with systemic risk based on capability benchmarks, safety benchmarks and reach (Article 51(1) and Annex XIII).

These studies reflect the outcome of the scientific and technical analysis of a series of external experts to the Commission. In some cases, they present a state-of-the-art review, while in others, they propose methodologies based on solid scientific evidence, while acknowledging significant uncertainty, as many of these problems still lack a widely accepted solution. The content, analysis, recommendations, and suggestions should then not be interpreted in any way as the position of the Commission, nor of the JRC editors in particular, but rather as the opinion of the authors.

Executive summary

The evaluation of systemic risks (SR) in General-Purpose AI (GPAI) models is a multifaceted challenge that extends beyond traditional capability assessments. This report analyses the role of AI safety benchmarks (or more precisely, risk and reliability benchmarks) as one possible tool that could inform the classification of GPAI models as GPAI models with systemic risk (GPAISR models). While most evaluations focus on measuring model capabilities, it is increasingly recognised that advanced capabilities do not necessarily equate to high risk, nor does a lack of frontier capabilities preclude a model from posing risk.

This document defines a framework that could inform the classification of GPAI models as GPAISR models, taking into account both their **advanced general capabilities** and their demonstrated **propensity for specific harms**, as identified through **targeted safety benchmarks**. This is part of a three-part recommendation that also includes capability-based and reach-based criteria presented in other reports of this collection.

The report details the current landscape of safety benchmarks, acknowledging their developmental stage and limitations, yet underscoring their unique ability to directly assess harmful outputs and behaviours. We propose a tiered evaluation strategy to manage cost and ensure proportionality, alongside a strong emphasis on a need for more standardisation, rigour, and the indispensable role of red-teaming (involving human experts).

1 Introduction

1.1 GPAISR Labelling and its Implications

The classification of general-purpose AI (GPAI) models as GPAI models with systemic risk (GPAISR models) is not merely a technical label but a trigger for more stringent testing, oversight, and compliance requirements, as stipulated by the EU AI Act (European Union, 2024).

Article 51 of the EU AI Act specifies that a GPAI model shall be classified as a GPAISR model if it (a) possesses high-impact capabilities, evaluated using appropriate technical tools and methodologies, or (b) if the Commission determines it has **equivalent capabilities or impact** based on criteria in Annex XIII, which includes benchmarks and evaluations (Independent Experts, 2025). Article 3 defines "systemic risk" as risks having a significant impact due to their reach or due to actual or reasonably foreseeable **negative effects on public health, safety, public security, fundamental rights, or society as a whole**. Article 55 requires that such models undergo rigorous scrutiny before and during their deployment.

Furthermore, the GPAI Code of Practice[1] defines 16 commitments related to safety and security for providers of GPAISR models, such as the adoption of a Safety and Security Framework (SSF) for risk assessment, mitigation, and governance, and rigorous analysis of identified systemic risks (Independent Experts, 2025). Hence, the GPAISR labelling process carries substantial economic and operational consequences for AI developers and deployers. An inaccurate or overly broad labelling mechanism could inadvertently stifle innovation or impose disproportionate burdens, particularly on smaller entities, potentially hindering the very progress it seeks to govern responsibly. Conversely, an overly narrow or incomplete labelling framework risks leaving significant systemic dangers unaddressed, exposing society to unforeseen harms.

1.2 Capability-Centric Evaluation and Limitations

Assessing model capabilities is a critical aspect of GPAISR assessment. The AI Act defines high-impact capabilities as those that match or exceed the capabilities recorded in the most advanced general-purpose AI models. These capabilities are strongly correlated with the potential for novel, complex, and potentially harmful emergent behaviours. Such behaviours may include sophisticated novel reasoning, advanced deception, or strategic behaviour.

However, a fundamental limitation of relying solely on capability measures is that "high capability" does not directly necessarily equate to "high risk". A model might demonstrate sub-frontier performance on general intelligence benchmarks, yet still have advanced reasoning or knowledge acquisition, and still exhibit dangerous propensities such as facilitating nuclear attacks or facilitating cyberattacks. These specific harmful behaviours are often not directly measured by general capability scores. Hence, we risk overlooking models that appear "safe" by general intelligence standards but are "risky" due to their specific propensities, deployment context, or scale of use. A more comprehensive assessment that directly probes for harmful behaviours and contextual vulnerabilities (e.g. safety benchmarks) would be highly beneficial as a measure to be considered, complementary to the other mechanisms contemplated.

Even then, fixed benchmarks may not be able to probe for unforeseen and complex harmful interactions that arise from advanced capabilities, or they may be circumvented by expert adversaries. Dynamic evaluation techniques, such as red-teaming and continuous post-market monitoring are needed to effectively monitor and mitigate risk.

[1] At the time of writing this document, the latest available version of the Code of Practice was the third draft (Independent Experts, 2025)

1.3 Sources and Scenarios of Systemic Risk

The EU GPAI Code of Practice identifies specific types of systemic risks that mandate continuous assessment (Independent Experts, 2025). These categories serve as useful guides for the development of targeted safety benchmarks. The primary types of systemic risk that must always be assessed include:

- **Cyber offence:** Risks associated with enabling large-scale, sophisticated cyber-attacks.
- **Chemical, biological, radiological and nuclear (CBRN) risks:** Risks of lowering barriers for malicious actors to enable CBRN attacks or accidents.
- **Harmful manipulation:** Risks of enabling targeted distortion of behaviour causing significant harm on a large scale, particularly through multi-turn interactions.
- **Loss of control:** Risks related to the inability to oversee and control autonomous GPAI models, potentially leading to large-scale safety or security threats.

Beyond these specific categories, the Code of Practice also identifies various sources of systemic risk, including "model propensities (e.g., misalignment, harmful tendencies, discriminatory bias)" and "model affordances/contextual factors (e.g., access to tools, scalability, release strategies, human oversight)" (Independent Experts, 2025). The identification of these concrete, high-consequence scenarios and their underlying sources means that systemic risks are not abstract concepts. This categorisation directly informs the design of highly specialised and context-aware safety benchmarks. For example, a general reasoning benchmark might not reveal a model's ability to synthesise dangerous biological information, but a targeted safety benchmark specifically designed for CBRN risks would. This requires that (a) "risk-specific" evaluations are developed, tailored to the identified systemic risk categories, and (b) a crucial need for domain expertise (e.g., cybersecurity experts for cyber offence benchmarks, biologists for CBRN benchmarks) in the development and implementation of these specialised safety evaluations.

2 The Role of Safety Benchmarks as a Systemic Risk Criterion

2.1 Current Landscape of Safety Benchmarks

While capability benchmarks, such as MMLU-Pro (Wang et al., 2024) or HumanEval (Chen et al., 2021), are relatively mature and widely adopted for assessing general intelligence, safety benchmarks are still under active development. Despite their nascent stage, these benchmarks are making progress in directly assessing AI risk and identifying specific harms that may not be evident through capability scores alone.

There is a wide range of safety evaluation datasets. For instance, SafetyPrompts (Röttger et al., 2025) is a survey that includes 144 evaluations covering many aspects of AI safety, including socio-demographic bias, toxic content generation, power-seeking, and sycophantic behaviours. Note that many of these benchmarks only target very specific aspects, and many contain model-written evaluations, which can be unreliable (Dev and Hobbhahn, 2024).

Much more general, open safety evaluation frameworks for GPAI include:

- **Inspect AI (UK AISI):** Developed by the UK AI Security Institute, Inspect AI is a framework for LLM evaluations that offers fine-grained assessments. Inspect supports a wide range of model providers and local models and includes various benchmarks, including cybersecurity and safeguards (UK AISI, 2025).
- **AI Luminate (MLCommons):** AI Luminate is an open-source framework specifically designed for safety evaluations, with community-created benchmarks covering 12 risks, including CBRNe (MLCommons, 2025). It assesses an AI model's or system's resistance to prompts designed to elicit dangerous, illegal, or undesirable behaviour across these hazard categories. It uses a mix of public and private prompts to prevent gaming.
- **LM Eval Harness (EleutherAI):** One of several general frameworks that allow any language model to be tested on the same inputs and codebase (Gao et al., 2024). Although not specifically designed for safety evaluation, several safety benchmarks have been developed on it, such as Phare (Giskard AI, 2025).

For specific systemic risk categories, several benchmarks are emerging:

— **Cybersecurity:** This is a quite mature field, with benchmarks that can measure LLM propensity to abuse code interpreters, offensive cybersecurity capabilities, susceptibility to prompt injection, visual prompt injection, spear phishing, and autonomous offensive cyber operations. For instance, Purple Llama CyberSecEval (Meta) provides a mature industry-wide set of cybersecurity safety evaluations based on industry standards like CWE and MITRE ATT&CK (Meta, 2025).

— **CBRN Risks:** Several efforts are underway to evaluate whether AI models and systems could enable the development of biological and chemical weapons, including through modified systems lacking safety guardrails (RAND, 2025). Anthropic's AI Safety Level (ASL) framework, for instance, includes deployment measures focused on preventing models from assisting with CBRN-weapons related tasks, particularly biological weapons (Anthropic, 2025).

— **Harmful Manipulation:** This area is in its early stages of evaluation. Examples include OpenAI's make-me-pay benchmark (OpenAI, 2025), where an AI attempts to persuade another AI to transfer money. And benchmarks like OpenDeception (Wu et al., 2025) exploring scenarios such as telecommunications fraud, product promotion, and emotional deception, where LLMs show high deception intention and success rates. Other relevant benchmarks include TruthfulQA (Lin et al., 2021), ToxiGen (Hartvigsen et al., 2022), ForbiddenQuestions (Shen et al., 2024), and VLDBench (Raza et al., 2025).

— **Loss of Control:** This area is also in its early stages of evaluation. The Time horizon metric, developed by the Model Evaluation & Threat Research (METR) team, measures the length of tasks AI agents can complete autonomously with 50% reliability, defined by the time a skilled human would take. This metric has shown exponential growth, doubling approximately every seven months since 2019, indicating rapid advancements in AI's ability to handle longer, more complex tasks (Kwa et al., 2025).

The current landscape of safety benchmarks, however, is not without its limitations. Many existing safety benchmarks are still in their early stages of development and offer limited coverage. For example, while AI Luminate measures certain risks effectively, it does not cover all potential harms. The GPAI Code of Practice's focus is also somewhat different from most traditional AI safety research, with more emphasis on security/misuse and less on content safety (e.g., bias), indicating that more research is needed to effectively cover the risks specified in the Code. GPAI providers are currently working on their own safety frameworks, such as Google Deepmind's Frontier Safety Framework, Anthropic's Responsible Scaling Policy, and OpenAI's Preparedness Framework, but there is a lack of common standards and open frameworks. This immaturity and incomplete coverage present a critical gap between the recognised need for direct safety assessment and the availability of comprehensive, robust tools.

Consequently, there is an urgent need for accelerated research and development in safety benchmarking. Policy initiatives should actively incentivise the creation of more comprehensive, robust, and standardised safety benchmarks, potentially through public-private partnerships or dedicated funding mechanisms, to close this critical assessment gap and enable more effective systemic risk identification.

2.2 How Safety Benchmarks Complement Capability-based Considerations for GPAISR Labelling

Safety benchmarks provide an essential layer of scrutiny that extends beyond general capabilities, offering a complement to capability-based considerations for GPAISR labelling. They possess the

ability to identify models that, even if not among the strongest in terms of general intelligence, nonetheless possess specific propensities for harm or can be easily misused to cause substantial damage.

Safety benchmarks are specifically designed to test for these "significant risks". They act as a "propensity filter", identifying models that, regardless of their general intelligence, exhibit dangerous characteristics relevant to specific systemic risk categories. This direct approach shifts the assessment paradigm from inferring potential harm based on a model's general capabilities to demonstrating actual harmful propensities through empirical testing. This empirical grounding can provide a concrete and actionable basis for regulatory action and, perhaps more importantly, the implementation of targeted mitigation strategies.

3 Practical Approaches to GPAISR Evaluation Using Safety Benchmarks

3.1 Integrating Safety Benchmarks

The GPAI Code of Practice explicitly supports a multi-faceted approach for systemic risk evaluation, emphasising a "multi-stage process" and a "variety of evaluation methods to ensure comprehensive and rigorous assessment" (Independent Experts, 2025). These methods include "Q&A sets, task-based evaluations, benchmarks, red-teaming, human uplift studies, and simulations" (Independent Experts, 2025). This guidance underscores that no single evaluation method is sufficient for a complete systemic risk assessment. For example, a model with high general capabilities, as indicated by strong performance on MMLU-Pro, might still require targeted safety benchmarks to uncover specific vulnerabilities related to CBRN risks. Furthermore, even with strong performance on both capability and safety benchmarks, emergent vulnerabilities that static benchmarks might miss could still be uncovered through dynamic red-teaming. This suggests that a simple numerical score will not suffice to establish a complete risk profile. That said, static benchmarks are already highlighting important problems with GPAI models. For instance, in the AI Luminate benchmark (MLCommons, 2025), many models have a failure rate higher than 10%. Regulatory frameworks should therefore encourage a flexible, adaptive portfolio of assessment techniques that can evolve with AI capabilities and the emerging risk landscape.

3.2 A Tiered Approach to Safety Benchmarking

Given that comprehensive safety benchmarking can be expensive and resource-intensive, a tiered approach is highly advisable to ensure efficiency and proportionality. This approach ensures that extensive and costly safety evaluations are primarily conducted for models already deemed sufficiently capable to represent a systemic risk, or those intended for deployment in high-risk domains.

A practical tiered strategy could involve:

- **Tier 1 (Initial Capability Screen):** All GPAI models would first undergo initial, relatively low-cost capability assessments, using a 'basket of benchmarks' approach based on observational scaling laws to measure general capabilities, as well as novel (yet efficient) state-of-the-art techniques explicitly measuring harmful manipulation and loss of control, which may involve (simulated) human-AI interaction, such as simulated user chats. Models that fail to meet a predefined minimum capability threshold and do not raise red flags regarding manipulation and loss of control, would generally be exempted from further extensive safety benchmarking, as their lower general capabilities would likely limit their potential for widespread systemic harm, unless they are specifically designed for or deployed in highly sensitive domains. This initial screen helps filter out models that are unlikely to pose any systemic risks.
- **Tier 2 (Targeted Safety Assessment):** Models that exceed the initial capability threshold, or those identified for use in high-risk domains regardless of their general capability level (e.g., medical diagnostics, critical infrastructure), would then proceed to more rigorous and targeted safety benchmarks. These benchmarks would focus on specific systemic risk categories, such as CBRN risks, cyber offence, or harmful manipulation. This ensures that models with the highest potential for impact are thoroughly vetted for harmful propensities and vulnerabilities.

This tiered strategy allows for focused resource allocation, directing more intensive scrutiny towards models that genuinely warrant it, while avoiding unnecessary overhead for less impactful models.

The minimum capability threshold or red flags regarding manipulation and loss of control should be determined by a multi-disciplinary expert group (also see section 4.3), based on progressive insight into AI risks and the availability of efficient benchmarking frameworks.

3.3 Standardisation in Safety Benchmarks

The GPAI Code of Practice (Measure II.4.5) explicitly mandates "Rigorous Model Evaluations" that meet "high scientific and technical rigour," ensuring "internal validity (accuracy in evaluation setting), external validity (generalisability to real-world contexts), and reproducibility (consistent results with same inputs and conditions)" (Independent Experts, 2025). The document also highlights a current "lack of standards and open frameworks" for safety evaluation, emphasising the need for more research, open frameworks, and SR evaluation standards.

Achieving this requires clear, standardised measurement procedures and robust mechanisms to prevent gaming or strategic underperformance by model providers. Without standardised procedures and robust anti-gaming measures, benchmark results can be manipulated, become incomparable across different models or providers, or fail to elicit a model's true harmful propensities, thereby undermining their utility for regulatory compliance and public trust[2]. We recommend that regulatory and policy bodies play an active role in defining and overseeing the development of safety benchmarks, similar to their proposed role for capability benchmarks.

[2] See (Eriksson et al., 2025) for an interdisciplinary review of current issues with AI benchmarks.

4 Policy Implications and Future Directions

4.1 Towards a Holistic Assessment Paradigm

There is some debate regarding the sufficiency of capability benchmarks versus the necessity (and maturity) of safety benchmarks for comprehensive risk assessment, but likely this is more accurately understood as a call for a holistic assessment paradigm. The optimal approach is likely not to choose one over the other but to recognise their distinct yet complementary roles. Capability benchmarks serve to identify models with high potential, indicating their capacity for advanced tasks and the scale of their potential impact. Safety benchmarks, conversely, directly assess harmful propensities, revealing a model's likelihood to generate dangerous content, exhibit biases, red flags regarding misdirection and loss of control, or be misused by rogue actors in specific high-risk scenarios. Both are important for a comprehensive and robust GPAISR labelling framework.

4.2 Addressing Cost and Resource Implications for Widespread Adoption

The cost of extensive benchmarking, particularly for safety evaluations, represents a significant practical barrier to widespread adoption and compliance. To overcome these cost implications, innovative strategies are required beyond individual provider efforts. The tiered approach discussed previously might offer a solution, applying more rigorous and costly safety evaluations only to models that have crossed a preliminary capability threshold or are designated for high-risk applications. Furthermore, shared benchmarking infrastructure could significantly reduce the burden on individual developers. Incentivising the development of open-source safety benchmarks, which are publicly available and easy to use (similar to the criteria for capability benchmarks), would also lower the barrier to entry for compliance.

4.3 Insights from the EU GPAI Code of Practice

The EU GPAI Code of Practice provides a robust and comprehensive framework for assessing and mitigating systemic risks associated with GPAI models, emphasising a multi-stage process, rigorous evaluations, and continuous monitoring. As safety benchmarks mature, they will play an increasingly important role in directly assessing harmful propensities. However, the rapid and continuous evolution of AI capabilities necessitates a dynamic and adaptable approach to benchmarks. Static safety benchmarks will become obsolete as models advance. The recommendation for the enforcement authorities to update capability benchmarks, thresholds, models, and measurement descriptions every six months should likely be extended to safety benchmarks. This could be realised by establishing multi-disciplinary benchmarking expert groups specifically dedicated to the continuous identification, development, and validation of new capability and safety benchmarks. Furthermore, policy would benefit from sufficient flexibility to explicitly allow evaluations to be replaced by better assessment techniques as they emerge, fostering innovation in safety measurement itself.

5 Conclusions and Final Recommendations

The evaluation of systemic risks in GPAI models would benefit from a sophisticated and comprehensive framework that goes beyond a singular focus on model capabilities. While capability benchmarks are valuable for identifying models with high potential impact, they are insufficient on their own to capture the full spectrum of systemic risks. Safety and reliability benchmarks are valuable complements, offering the unique ability to directly probe for harmful propensities, biases, and vulnerabilities, regardless of a model's general intelligence. This dual-trigger approach is important for accurate GPAISR labelling, ensuring that models posing significant risks are subjected to the necessary rigorous scrutiny.

The current landscape of safety benchmarks is still in development but demonstrates potential to provide empirical evidence of a model's "risk surface." However, their current immaturity, limited coverage, and potential cost necessitate strategic implementation, combined with robust standardisation and increased incentives to establish robust safety benchmarks. At present, we can make the following recommendations:

- **A Dual-Trigger GPAISR Labelling Framework:** Envision a system where models can be classified as GPAISR models based on a combination of their high-impact general capabilities (as assessed by capability benchmarks) and their demonstrated harmful propensities (identified through safety benchmarks).
- **Prioritise Incentives for Safety Benchmark Development:** Fund and incentivise the rapid creation of a diverse, comprehensive, and robust suite of safety benchmarks. These benchmarks should cover the key identified systemic risk categories (e.g., CBRN, cyber offence, harmful manipulation, and loss of control). Foster public-private collaboration and open-source initiatives to accelerate this development.
- **Implement a Tiered Evaluation Strategy:** Design a cost-effective tiered approach where initial, less expensive capability assessments serve as a preliminary filter. More rigorous and targeted safety benchmarks should then be applied only to models that meet a predefined capability threshold or are intended for deployment in high-risk domains, ensuring proportionality and efficient resource allocation.
- **Standardise Rigorous Evaluation Protocols:** Establish clear, reproducible procedures for running all benchmarks. There is a need for more research, open frameworks, and SR evaluation standards.
- **Establish a Dynamic Benchmark Governance Mechanism:** Create an agile governance body, such as a multi-disciplinary expert group, responsible for the continuous review, update, and phasing in/out of both capability and safety benchmarks. This ensures that evaluation methods remain relevant, address emerging risks, and adapt to rapid advancements in AI.

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List of abbreviations and definitions

AI Artificial Intelligence

AISI Artificial Intelligence Security Institute

ASL AI Safety Level

CBRN Chemical, Biological, Radiological and Nuclear

CoP Code of Practice

CWE Common Weakness Enumeration

GPAI General-Purpose Artificial Intelligence

GPAI models General-Purpose Artificial Intelligence models

GPAISRs/GPAISR/GPAI-SR General-Purpose Artificial Intelligence models with Systemic Risk

METR Model Evaluation & Threat Research

SR Systemic Risk

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